Talking about motion

LEARNING TARGET · K1 I can explain how position, distance, and displacement differ.

SECTION A KEY VOCABULARY

Position	*Where are you?* Measured from a chosen zero, with a chosen + direction.
Distance	*How much travel?* Counts every part of the path. Never negative.
Displacement	*What changed from start to finish?* The sign gives the direction.
Scalar / vector	Distance has size only (scalar). Displacement has size *and* direction (vector).

SECTION B KEY EQUATIONS + UNITS

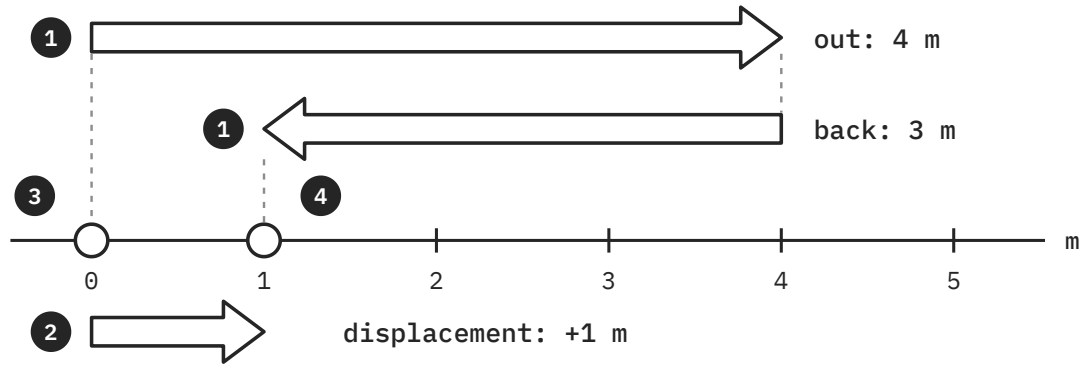
$$\Delta x = x_f - x_i$$

distance = leg 1 + leg 2 + …

- Δ "change in" (finish − start)
- x position, meters (m)

SECTION C COLOR IT · ONE TRIP, THREE QUESTIONS

0 m → 4 m → 1 m. Right is +.



- ☐ **1 · PENCIL** Distance: trace both legs
- ☐ **2 · BLUE** Displacement arrow
- ☐ **3 · BLUE** Start position
- ☐ **4 · BLUE** Finish position

SECTION D WORKED EXAMPLE

A cart goes from 2 m to 6 m, then back to 4 m. Right is positive. Find the distance and the displacement.

GIVEN	EQUATION	WORK	ANSWER
$x_i = 2$ m turn at 6 m $x_f = 4$ m	distance = leg 1 + leg 2 $\Delta x = x_f - x_i$	4 m + 2 m 4 m − 2 m	distance **6 m** displacement **+2 m**

SECTION E THE 2026-XJ THREAD

DAY 2 OF 22 · NEW TOOL: A REFERENCE FRAME

Before we can say where XJ is, we need a zero and a + direction. Analysts measure its position from Earth and track its displacement, not the winding path it travels.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Where you finish is not the whole trip

Position, distance, and displacement

If I say a cart is at two meters, you need to know where we started measuring. Position needs a chosen zero, called the origin. We also choose a positive direction. On today's notes page, right is positive. Once those choices are clear, a position tells us where the cart is along the line.

Position is a location. Distance counts how much travel occurred. Displacement describes the change from starting position to final position. Those are three different questions, so one trip can have three different descriptions. I want you to decide which question is being asked before calculating anything.

Trace both legs in **Section C**. It shows a trip from **0 m to 4 m, then back to 1 m**. The outward leg is four meters long. The return leg is three meters long. Total distance is **distance = 4 m + 3 m = 7 m**. Coming back does not erase the first leg. Distance counts every part of the route and is never negative.

Displacement uses only the starting and finishing positions. The symbol Δ means "change in," so Δx means change in position. The rule is $\boldsymbol{\Delta x = x_f - x_i}$: finish minus start. For the diagram, $\boldsymbol{\Delta x}$ **= 1 m − 0 m = +1 m**. The cart finishes one meter right of where it began, even though it traveled seven meters.

Now find the starting and turning positions in **Section D**. The cart starts at **2 m**, travels to **6 m**, and returns to **4 m**. Its first leg is **6 m − 2 m = 4 m**. The return leg has length **6 m − 4 m = 2 m**. Add the lengths to find **distance = 4 m + 2 m = 6 m**. The direction of a leg does not make its length negative.

For displacement, use the endpoints instead: $\boldsymbol{\Delta x}$ **= 4 m − 2 m = +2 m**. The final position is 4 m from our chosen zero, while the displacement is 2 m right of the starting point. Those answers differ because the cart did not start at zero. This is a common place to lose track of what a number means.

Distance is a scalar: it has size and units. Displacement is a vector: it also has direction. Along one axis, the sign gives that direction. A negative displacement would mean a finish left of the start. In the diagram, the traveled route stays in pencil; the positions and displacement arrow are blue.

For our fictional asteroid, 2026-XJ, we need the same clear reference frame. Measuring from Earth gives our classroom model a zero, but we must still name the positive direction. Total distance describes the route traveled. Starting position and displacement tell us where the asteroid ends up. That is the information a position forecast needs.